Question	Answer	Marks
1(a)(i)	$v_H = 28 \times \cos 34°$ $= 23\ \text{m s}^{-1}$	**A1**
	$v_V = 28 \times \sin 34°$ $= 16\ \text{m s}^{-1}$	**A1**
1(a)(ii)	time = 16 / 9.81 = 1.6 s	**A1**
1(a)(iii)	horizontal straight line at $v = 23\ \text{m s}^{-1}$ from $t = 0$ to $t = 3.2$ s	**B1**
1(a)(iv)	straight diagonal line starting at a positive velocity from $t = 0$ to $t = 3.2$ s, crossing the time axis	**B1**
	line starting at $v = 16\ \text{m s}^{-1}$ and ending at $v = -16\ \text{m s}^{-1}$	**B1**
	line passing through $v = 0$ at $t = 1.6$ s	**B1**
1(b)(i)	product of mass and velocity	**B1**
1(b)(ii)	$F = \Delta p / t$	**C1**
	$= 13 / 3.2$ $= 4.1$ N	**A1**

Question	Answer	Marks
1(b)(iii)	$m = \Delta p / v$ $= 13 / (2 \times 16)$ $= 0.41$ kg **or** $m = F / g$ $= 4.06 / 9.81$ $= 0.41$ kg	**A1**

Question	Answer	Marks